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CCRDet: Cross-modal complementary region-aware framework for drone-based RGBT tiny person detection

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ABSTRACT

Drone-based RGBT person detection, which has garnered significant attention, seamlessly integrates the spatial flexibility of drones with the around-the-clock time flexibility of RGBT data for continuous information acquisition. However, the drone-captured images cover a wide range of areas, resulting in illumination imbalance and thermal background clutter issues in RGB and thermal images. The varying quality of the two modalities in different regions of an RGBT image set increases the difficulty of complementary information fusion, thereby leading to false negatives in the advanced drone-based RGBT tiny person detection methods. In this context, a novel Cross-modal Complementary Region-aware framework for drone-based RGBT tiny person Detection (CCRDet) is proposed for effective object detection under poorly illuminated RGB regions and thermally cluttered backgrounds. CCRDet employs the proposed cross-modal region-aware guidance to be aware of these regions and guide the counterpart modality to enhance valid target features accordingly. After that, it leverages the proposed modality-difference feature gated fusion to deliver these valid target features to the fused features with effective preservation, thereby enhancing their response intensity after fusion and providing high-quality inputs for the detection head. Extensive experiments on two drone-based RGBT tiny person detection datasets, RGBT-DronePerson and VTUAV-det, demonstrate the effectiveness of the proposed method. The code is available at <https://github.com/G-pz/CCRDet>.

1. Introduction

Person detection is a crucial area of computer vision, with various applications such as surveillance, traffic monitoring, and search & rescue [1]. Drone-based imagery can capture distant scenes and provide wide-range bird's-eye views, thereby offering spatial flexibility for pedestrian detection [2]. In parallel, thermal cameras can capture thermal radiation emitted by objects, thereby facilitating pedestrian detection under low-light conditions and ensuring temporal flexibility across varying illumination scenarios [3]. Consequently, drone-based RGBT tiny person detection has garnered significant attention, as it affords remarkable flexibility across both temporal and spatial domains.

Current drone-based RGBT tiny person detection methods primarily adopt mid-fusion strategies based on deep models, which have become the mainstream paradigm [4,5]. As illustrated in Fig. 2(a), these methods extract features from RGB and thermal modalities using separate backbones, and then fuse them at intermediate network layers. Nonetheless, in challenging scenarios such as poorly illuminated regions in RGB images and background clutter in thermal images, advanced methods may face performance limitations. As illustrated in Fig. 1(a), pedestrians

are indistinguishable in the poorly illuminated regions, while Fig. 1(b) shows that thermal background clutter obscures pedestrian signatures. Consequently, in the fusion-based detection results of the RGBT image sets produced by the advanced method QFDet [5], multiple instances of false negatives are observed. The drone-captured images cover a wide range of areas, resulting in illumination imbalance and thermal background clutter issues in RGB and thermal images [5]. Consequently, the varying quality of the two modalities in different regions of an RGBT image set increases the difficulty of complementary information fusion.

To address the issues and limitations above, a novel Cross-modal Complementary Region-aware framework for drone-based RGBT tiny person Detection (CCRDet) is proposed for effective object detection under poorly illuminated RGB regions and thermally cluttered backgrounds. As illustrated in Fig. 2(b), the proposed method is based on the mid-fusion scheme, including cross-modal region-aware guidance (CRG) and modality-difference feature gated fusion (MFGF). To effectively leverage complementary information at locations corresponding to the aforementioned regions, the CCRDet employs CRG to be aware of these regions and guide the counterpart modality to enhance valid target features in the corresponding complementary areas. Specifically,

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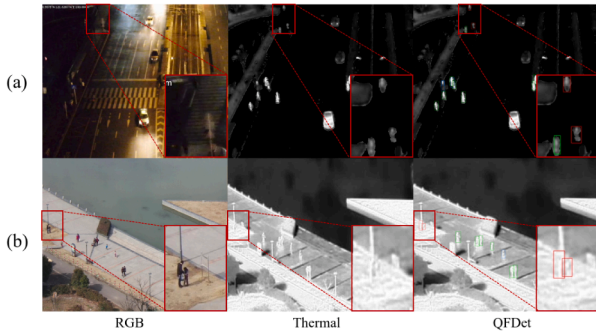


Fig. 1. Illustrative example of an advanced drone-based RGBT tiny person detection method. (a) Poorly illuminated regions in RGB images lead to false negatives (red boxes) in fusion-based results of QFDet. (b) Thermal background clutter causes false negatives (red boxes) in fusion-based results of QFDet.

the CRG addresses low-light regions and background clutter by introducing the Luminance Perception Module (LPM) and the Background Clutter Attention Module (BCAM), respectively. Since the Luminance Map (L-Map), obtained via RGB-to-YUV conversion with subsequent extraction of the Y channel, provides an independent representation of image luminance [6], the LPM generates a low-light region-aware map by performing luminance perception through the joint utilization of the luminance map and RGB features. Meanwhile, the large receptive field excels at aggregating contextual information across a broader spatial range, enabling effective perception of large-scale, highly homogeneous regions [7]. The BCAM generates a background clutter-aware map by applying background clutter attention to filter thermal features with a large receptive field. Then, the CRG leverages these region-aware maps to cross-guide each modality to focus on these complementary regions and enhance valid target features accordingly. After that, the MFGF is proposed to enhance the feature response intensity of these valid target features after fusion. Subtraction-based modality-difference features facilitate estimating whether valid target features exist in each modality per region, and gate the dominant modality accordingly. It delivers valid target features to the fused features with effective preservation, thereby enhancing their response intensity after fusion.

The contribution of this work can be summarized as follows.

(1) A novel CCRDet framework is proposed to address the false negatives caused by poorly illuminated regions in RGB and background clutter regions in thermal, including cross-modal region-aware guidance and modality-difference feature gated fusion. The CCRDet employs CRG and MFGF to enhance valid target responses in the aforementioned counterpart regions and deliver them to the fused features with effective preservation. Finally, the well-fused features are fed into a detection head for classification and regression.

(2) To effectively leverage complementary information at locations corresponding to poorly illuminated regions in RGB and background clutter regions in thermal, the CRG is proposed. It employs the LPM with explicit Y-channel extraction and the BCAM with large-receptive-field-based filtering to generate Low-Light Region-aware and Background Clutter Region-aware maps (LLR-Map and BCR-Map), which guide each modality toward the uninformative regions of its counterpart, thereby enhancing valid target features in the complementary regions.

(3) To enhance the feature response intensity of valid target features after fusion, the MFGF is proposed. Subtraction-based modality-difference features facilitate estimating whether valid target features exist in each modality per region, and gate the dominant modality accordingly. It delivers valid target features to the fused features with effective preservation, thereby enhancing their response intensity after fusion.

Extensive experiments are conducted on the RGBTDronePerson and VTUAV-det datasets to evaluate the effectiveness of CCRDet. Comparative and visualization experiments demonstrate that CCRDet achieves

competitive performance. Feature comparison and analysis intuitively verify that CCRDet is effectively aware of poorly illuminated regions and background clutter regions, thereby alleviating the false negatives caused by these regions.

The rest of this paper is organized as follows: In Section 2, the related works are briefly described. Section 3 gives a detailed description of CCRDet. Then, Section 4 validates the effectiveness of the proposed method and discusses the experimental results. Finally, the conclusion of this paper is made in Section 5.

2. Related work

2.1. Drone-based tiny object detection

Improvements for drone-based tiny object detection encompass a variety of strategies, among which the following are commonly adopted. The first line of research emphasizes multi-scale feature learning, aiming to capture discriminative information across different resolutions. Since tiny objects in aerial imagery often lack sufficient semantic cues, feature pyramid networks [8] and path aggregation networks [9] have been widely used to integrate low-level spatial details with high-level semantics, thereby improving detection robustness across diverse target scales. Multi-scale feature fusion [10] and adaptive channel weighting mechanisms have been shown to enhance small-object representations and reduce information loss across layers.

The second direction focuses on resolution enhancement, which seeks to address the limited pixel information of tiny instances. Techniques such as image-level or feature-level super-resolution [11], deconvolution [12], and region zoom-in [13] have been applied to recover fine-grained details, improving small-object visibility and detection accuracy. The third direction involves detection head refinement, where high-resolution detection layers [14], dense anchor designs [15], and scale-adaptive receptive fields [16] are introduced to better localize tiny instances, while pruning or simplifying layers intended for large objects to improve computational efficiency. The fourth direction explores data augmentation strategies, including mosaic or mixup operations, copy-paste of small objects, and adaptive cropping or scaling to simulate diverse UAV viewpoints [17]. These strategies enhance the diversity of small-object samples in training and improve model robustness without adding inference overhead.

However, these methods are primarily designed for visible (RGB) images and are insufficient to address the challenges posed by low-light conditions, such as nighttime scenarios.

2.2. RGBT fusion-based object detection

To overcome the limitations of single-modality detection, a large body of research has focused on fusion-based methods that integrate information from both visible and thermal images. According to the stage at which fusion occurs, existing approaches can be broadly divided into three categories: early fusion, late fusion, and mid fusion [18].

Early fusion combines RGB and thermal inputs at the pixel level or in shallow feature space [19]. This straightforward strategy allows the network to process both modalities simultaneously from the beginning, but it often fails to account for modality heterogeneity and is thus prone to introducing redundant or noisy information [20].

Late fusion performs detection independently in each modality and subsequently merges the outputs at the decision level [21]. Such approaches are relatively robust to modality-specific degradation since each detector can operate autonomously. However, they typically lack fine-grained cross-modal collaboration, which constrains their ability to exploit complementary cues effectively [20].

Mid fusion, by contrast, has become the dominant paradigm due to its superior detection performance [22]. It integrates features from both modalities at intermediate network layers, enabling more nuanced interactions and joint representations [18]. Various mid-fusion strategies

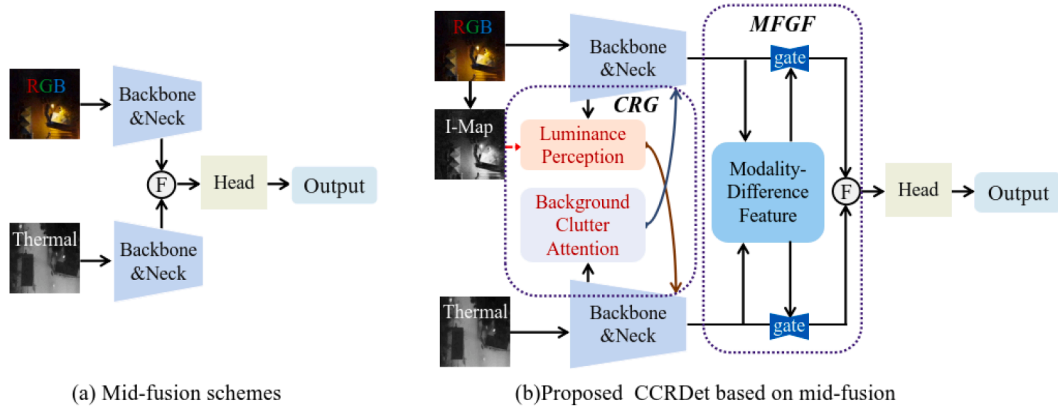


Fig. 2. Conceptual diagrams of (a) mid-fusion scheme in current methods, as well as (b) the designed CCRDet based on the mid-fusion scheme, including cross-modal region-aware guidance (CRG) and modality-difference feature gated fusion (MFGF).

have been proposed, including concatenation, element-wise addition, cross domain learning [23,24], and cross-attention mechanisms [25,26]. In addition, some luminance-aware approaches [22,27] have been introduced to predict weights for the entire visible image. These methods typically depend on implicit estimations rather than direct luminance sources and adopt a global weighting strategy that ignores local brightness differences across regions.

Nevertheless, effectively extracting and fusing features from heterogeneous modalities remains a formidable challenge, particularly in scenarios with varying illumination and thermal conditions.

Another approach employs RGB-to-LAB conversion for luminance perception [28]. However, the Y channel in YUV, using the linear formula $Y = 0.299R + 0.587G + 0.114B$, is favored for luminance representation due to its computational efficiency and stability [29], especially in challenging lighting conditions. Unlike the L channel in CIELAB, which employs a more complex, non-linear transformation [30], the Y channel's luminance-like nature and robustness make it a superior choice for multi-modal fusion, particularly when addressing illumination variations and prioritizing computational efficiency.

3. Method

The CCRDet is proposed for effective object detection under poorly illuminated RGB regions and thermally cluttered backgrounds. It employs the proposed cross-modal region-aware guidance (CRG) to be aware of these regions and guide the counterpart modality to enhance valid target features. After that, it leverages the proposed modality-difference feature gated fusion (MFGF) to deliver valid target features to the fused features with effective preservation, thereby enhancing their response intensity after fusion.

3.1. Overall architecture of CCRDet

As illustrated in Fig. 3, the architecture of the proposed CCRDet is constructed on a one-stage detection framework. To address the false negatives caused by poorly illuminated regions in RGB and background clutter regions in thermal, two novel methods are proposed for the heterogeneous feature extraction stage and the feature fusion stage. Specifically, in the heterogeneous feature extraction stage, the CCRDet employs CRG after each stage of the backbone to enhance valid target features in the aforementioned regions. Then, in the feature fusion stage, the CCRDet leverages MFGF to deliver valid target features to the fused features with effective preservation.

In this framework, ResNet50 [31] is utilized as the backbone network for both branches. The input of each Luminance Perception Module (LPM) is the output of the corresponding stages of the RGB modality

and the luminance map, obtained via RGB-to-YUV conversion with subsequent extraction of the Y channel. Then, the low-light region-aware map (LLR-Map) generated by the LPM is multiplied with the opposite thermal backbone networks to enhance valid target features in the poorly illuminated regions of the thermal modality. The input of each Background Clutter Attention Module (BCAM) is the output of the corresponding stages of the thermal modality. Then, the background clutter region-aware map (BCR-Map) generated by the BCAM is multiplied with the opposite RGB backbone networks to enhance valid target features in the background clutter regions of the RGB modality.

After that, the multi-level features of each modality are fused through a top-down feature pyramid network (FPN) [8]. Then, the MFGF is employed to fuse the features of the two modalities at each level of the FPN. Finally, the fused multi-level features are fed into the detection head for classification and regression.

3.2. Cross-modal region-aware guidance

To effectively leverage complementary information at locations corresponding to poorly illuminated regions in RGB and background clutter regions in thermal, the CRG is proposed. It employs the LPM with explicit Y-channel extraction and the BCAM with large-receptive-field-based filtering to generate Low-Light Region-aware and Background Clutter Region-aware maps (LLR-Map and BCR-Map), which guide each modality toward the uninformative regions of its counterpart, thereby enhancing valid target features in the complementary regions.

In detail, the LPM is designed to be aware of poorly illuminated regions in RGB images through luminance perception. Its inputs are features, which can be represented as F_{RGB} , out of the corresponding stages of the RGB modality and the Luminance Map. The LPM first downsamples the Luminance Map to the same size as the RGB features, and then concatenates the convoluted features along the channel dimension. The concatenated feature $\hat{F}_{RGB}$ can be formulated as:

$$\hat{F}_{RGB} = \text{Concat}(\text{Conv}(F_{RGB}), \text{Conv}(DS(\text{LuminanceMap}))), \quad (1)$$

where $DS(\cdot)$ denotes the downsampling operation, and $\text{Concat}(\cdot)$ indicates the concatenation operation. After that, a convolutional layer is employed to extract the LLR-Map from the concatenated features, which can be formulated as:

$$\text{LLR-Map} = \text{Sigmoid}(\text{ReLU}(\text{Conv}(\hat{F}))). \quad (2)$$

Finally, the LLR-Map is multiplied with the opposite thermal backbone networks to enhance valid target features in the poorly illuminated regions of the thermal modality.

Similarly, the BCAM is designed to be aware of background clutter regions in thermal images. Its input is features, which can be

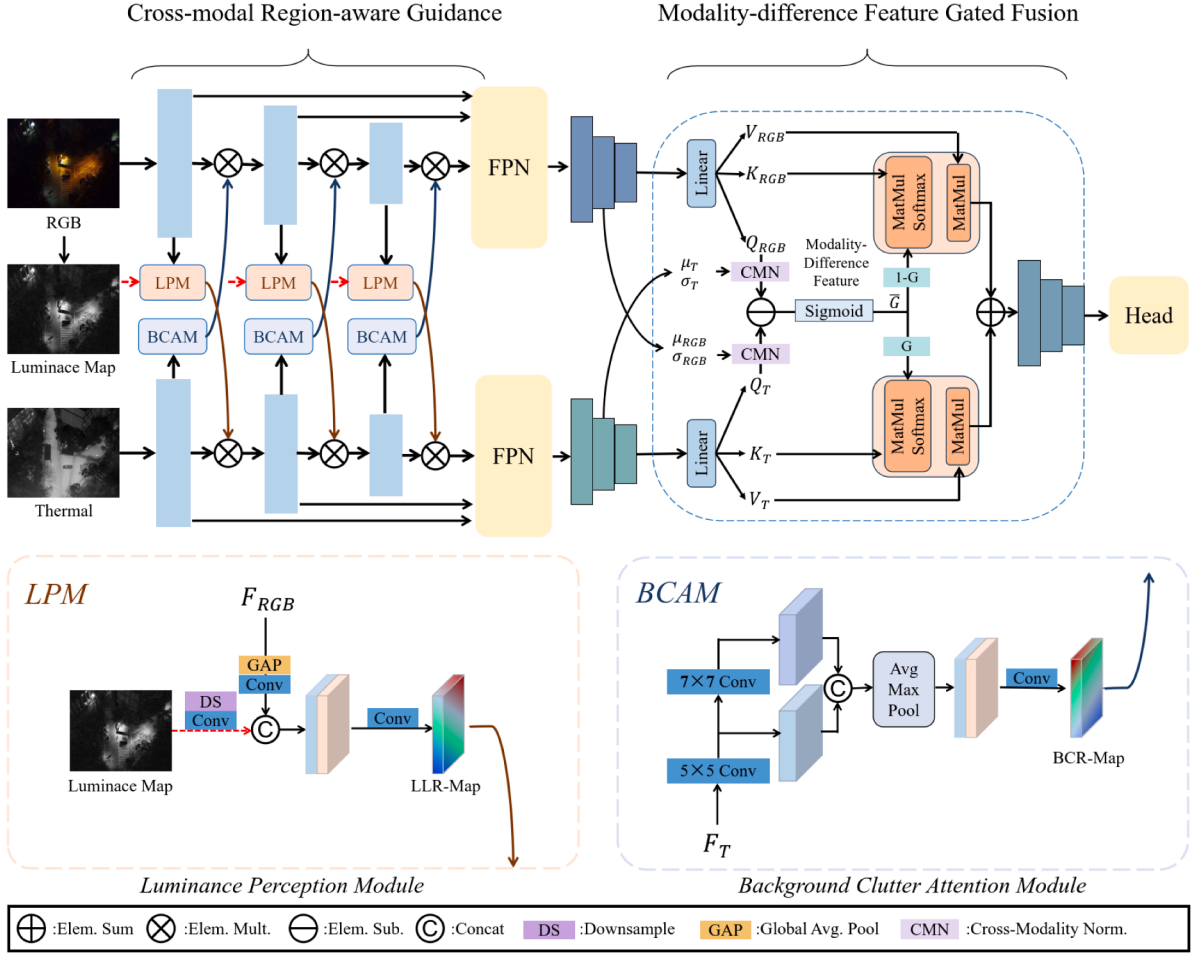


Fig. 3. The overall framework of the proposed method, CCRDet, is designed to address the false negatives caused by poorly illuminated regions in RGB and background clutter regions in thermal. It consists of two main components: Cross-modal Region-aware Guidance and Modality-difference Feature Gated Fusion.

represented as F_T , out of the corresponding stages of the thermal modality. The BCAM employs two big kernel convolutional layers to filter the background clutter information and concatenates the filtered features, which can be formulated as:

$$\hat{F}_T = \text{Concat}(\text{Conv}_{k=5}(F_T), \text{Conv}_{k=7}(\text{Conv}_{k=5}(F_T))), \quad (3)$$

where $\text{Conv}_{k=5}(\cdot)$ and $\text{Conv}_{k=7}(\cdot)$ denote the convolutional layers with kernel sizes of 5 and 7, and the strides are both set to 1 to maintain the spatial size. Their large receptive fields facilitate capturing extensive background clutter information. After that, a convolutional layer is employed to extract the BCR-Map from the concatenated features, which can be formulated as:

$$\text{BCR-Map} = \text{Sigmoid}(\text{ReLU}(\text{Conv}(\hat{F}))). \quad (4)$$

Finally, the BCR-Map is multiplied with the opposite RGB backbone networks to enhance valid target features in the background clutter regions of the RGB modality.

3.3. Modality-difference feature gated fusion

To enhance the feature response intensity of valid target features after fusion, the MFGF is proposed. Subtraction-based modality-difference features facilitate estimating whether valid target features exist in each modality per region, and gate the dominant modality accordingly. It delivers valid target features to the fused features with effective preservation, thereby enhancing their response intensity after fusion.

In detail, the MFGF first performs a linear projection of the features from both modalities to get feature descriptors. The generation process

is defined as:

$$\begin{aligned} Q_{RGB}, K_{RGB}, V_{RGB} &= \text{Split}(\text{Linear}(F_{RGB})), \\ Q_T, K_T, V_T &= \text{Split}(\text{Linear}(F_T)), \end{aligned} \quad (5)$$

where $\text{Linear}(\cdot)$ denotes the linear projection operation, and $\text{Split}(\cdot)$ indicates the splitting operation. Since there is modality discrepancy between the two images, directly subtracting feature descriptors of the two modalities is not optimal. Therefore, cross-modality normalization [32] is employed to alleviate the modality discrepancy. Then, the modality-difference features can be formulated as:

$$F_D = \text{Norm}(Q_{RGB}, \mu_T, \sigma_T) - \text{Norm}(Q_T, \mu_{RGB}, \sigma_{RGB}), \quad (6)$$

where $\text{Norm}(\cdot)$ denotes the normalization operation, and μ and σ represent the mean and standard deviation of the two modalities' features, respectively. After that, the gating coefficients of the two modalities are estimated based on the modality-difference features, which can be formulated as:

$$\begin{aligned} w_1 &= \frac{\exp(\text{clip}(F_D, \max_{50}))}{\exp(\text{clip}(F_D, \max_{50})) + \exp(\text{clip}(-F_D, \max_{50}))}, \\ w_2 &= 1 - w_1 \end{aligned} \quad (7)$$

where w_1 and w_2 indicate the gating coefficients of Q_{RGB} and Q_T , respectively. Clip operation with a maximum value is employed to avoid numerical overflow during the exponential operation. Then, the fused

feature descriptors can be formulated as:

$$F_{fused} = \text{Softmax}\left(\frac{\hat{Q}_{RGB} K_{RGB}^T}{\sqrt{d_k}}\right) V_{RGB} + \text{Softmax}\left(\frac{\hat{Q}_T K_T^T}{\sqrt{d_k}}\right) V_T, \quad (8)$$

where $\hat{Q}_{RGB} = w_1 Q_{RGB}$ and $\hat{Q}_T = w_2 Q_T$, d_k is the dimension of K_{RGB} and K_T . Finally, the fused features are fed into the detection head for classification and regression.

4. Experiments

4.1. Datasets

The proposed CCRDet was evaluated over two drone-based RGBT tiny person detection datasets involving RGB and thermal modalities, which are RGBTDronePerson [5] and VTUAV-det [5,33].

RGBTDronePerson. The RGBTDronePerson dataset is a large-scale drone-based RGBT tiny person detection benchmark. It contains 6125 pairs of RGBT images, each with a resolution of 640×512 pixels. These image pairs are collected across daytime and nighttime, spanning from winter to summer, to cover diverse lighting and temperature conditions. In addition, it offers extensive annotations featuring bounding boxes for four categories: person, rider, crowd, and uncertain. A total of 43,006 instances are annotated as “person,” 4869 instances as “rider,” and 7316 instances as “crowd”. 98.0% of these instances are smaller than 20 pixels.

The dataset is randomly divided into training and validation sets. Finally, the training set consists of 4900 pairs, and the validation set consists of 1225 pairs as well as [5]. The detection performance is evaluated on the validation set using mean Average Precision (mAP) [34] as the evaluation criterion.

VTUAV-det. The VTUAV-det is collected from VTUAV [33], which is a large-scale benchmark for visible-thermal UAV tracking, containing a variety of scenes and weather conditions. Finally, the VTUAV-det dataset consists of 16,770 pairs of RGBT images, each with a resolution of 1920×1024 pixels. Differently, this dataset only considers the “person” category. Besides, it presents a more significant scale variation. According to the COCO evaluation protocol [34], object sizes are categorized into “small”, “medium”, and “large”. The VTUAV-det training set has 7651 small objects, 50,794 medium objects, and 17,869 large objects. The testing set has 9864 small objects, 32,731 medium objects, and 5960 large objects.

The dataset is divided into training and testing sets following the original VTUAV dataset. Finally, the training set consists of 11,392 pairs, and the testing set consists of 5378 pairs as well as [5]. The detection performance is evaluated on the testing set using mean Average Precision (mAP) [34] as the evaluation criterion.

4.2. Implementation details

Training and Testing Details. The proposed CCRDet was implemented with the PyTorch [35] and MMDetectionV2 framework. In CCRDet, two ResNet-50 [31] pre-trained on ImageNet [36] were utilized as the backbones for the RGB and thermal modalities, respectively. The model was optimized using the SGD algorithm with momentum of 0.9 and weight decay of 0.0001. The number of epochs and batch size were set to 12 and 8, as well as [5]. For RGBTDronePerson, the image inputs were resized to 640×512 , the re-balance weights were all set to 1, and the IoU threshold for test time NMS was set to 0.3. For VTUAV-det, the image inputs were resized to 800×512 , the re-balance weights were set to linearly grow from 1 to 5, and the IoU threshold for test time NMS was set to 0.5. To avoid overfitting, the epoch with the best testing results was taken as the final model. Data augmentation included random rotation and random horizontal flips. The confidence score threshold is

set to 0.05. Experiments were carried out on a single NVIDIA RTX 8000 GPU (48 GB).

Loss function. Following the design of Generalized Focal Loss (GFL) [37], the detection head jointly optimizes classification confidence and localization quality. The overall training objective consists of three components: Quality Focal Loss (QFL) for classification, Distribution Focal Loss (DFL) for discrete localization modeling, and GIoU Loss for bounding box regression. The total loss is defined as:

$$\mathcal{L} = \mathcal{L}_{QFL}(p, y) + \lambda_{DFL} \mathcal{L}_{DFL}(d, \hat{d}) + \lambda_{GIoU} (1 - \text{GIoU}(B_p, B_{gt})). \quad (9)$$

Quality Focal Loss (QFL) supervises the classification score p with an IoU-aware soft label y , aligning classification confidence with localization quality. Distribution Focal Loss (DFL) models each bounding box side as a discrete distribution and refines localization through distribution learning. GIoU loss further optimizes the spatial overlap between predicted box B_p and ground-truth box B_{gt} , improving regression robustness. λ_{DFL} and λ_{GIoU} are the balancing weights for the DFL and GIoU Loss, respectively. In the experiments, λ_{DFL} and λ_{GIoU} were set to 0.25 and 2.0, respectively, as well as [37].

Baseline detectors customized for RGBT object detection. For comparison, several state-of-the-art RGBT object detectors were implemented as baselines. They were constructed with two-stream structures, where RGB and thermal features were extracted by separate backbones and fused via concatenation with a 1×1 convolution. GFL [37] was finally selected as the strong baseline due to its favorable trade-off between accuracy and speed on tiny objects.

4.3. Comparison with state-of-the-art methods

This section compares CCRDet with baseline detectors customized for RGBT object detection: Faster R-CNN [38], Cascade R-CNN [39], RetinaNet [40], FCOS [41], ATSS [42], GFL [37], the advanced tiny object detection algorithms: SSPNet [43], ATSS w/NWD [44], FCOS w/RFLA [45], and the advanced RGBT fusion-based object detection algorithms: CFT [46], HRFuser [47], TInet [4], QFdet [5], VTSaRNet [48], and IM-CMDet [49]. Experiments are conducted on the RGBTDronePerson and VTUAV-det datasets. The quantitative results are shown in Tables 1 and 2. These three types of methods correspond to the first, second, and third sections of the two tables, respectively. The visualization results are shown in Figs. 4 and 5.

On RGBTDronePerson. As shown in the first row of Table 1, the four size categories were defined as follows: tiny1 ([0, 8), 7.8%), tiny2 ([9, 12), 54.8%), tiny3 ([12, 20), 35.4%), and small ([20, 32], 1.9%). The category “tiny” was defined as the union of tiny1-3. As shown in Table 1 columns 3 and 4, the proposed CCRDet achieved an $\text{mAP}_{50}^{\text{all}}$ and $\text{mAP}_{50}^{\text{tiny}}$ of 43.58% and 45.14%. Compared to these representative methods, CCRDet achieves substantial improvements with increases of 1.67% and 1.42% in $\text{mAP}_{50}^{\text{all}}$ and $\text{mAP}_{50}^{\text{tiny}}$, respectively. These results demonstrate the effectiveness of the proposed CCRDet in drone-based RGBT tiny person detection.

Table 1 rows 2–6 show the performance of the baseline detectors customized for RGBT object detection. Among them, GFL achieved the best performance with an $\text{mAP}_{50}^{\text{all}}$ of 39.04% and an $\text{mAP}_{50}^{\text{tiny}}$ of 40.83%. Built upon the strong baseline GFL, the proposed CCRDet achieves improvements of 4.54% and 4.31% in $\text{mAP}_{50}^{\text{all}}$ and $\text{mAP}_{50}^{\text{tiny}}$, demonstrating the effectiveness of the proposed methods. Table 1 rows 7–10 show the performance of the advanced tiny object detection algorithms. The proposed CCRDet outperforms FCOS w/RFLA by 5.07% and 6.02% in $\text{mAP}_{50}^{\text{all}}$ and $\text{mAP}_{50}^{\text{tiny}}$. The result verifies the significance of employing effective fusion strategies for tiny object detection. Table 1 rows 11–14 show the performance of the advanced RGBT fusion-based object detection algorithms. HRFuser’s Transformer struggles with low-pass filtering for tiny objects [50], while TInet’s global illumination estimate fails in complex lighting, degrading tiny-target fusion in both. Among them, QFdet achieved the best performance with an $\text{mAP}_{50}^{\text{all}}$ of 39.61% and

Table 1

Performance comparison with the advanced methods on RGBTDronePerson. The four size categories are defined as follows: tiny1 ([0, 8), 7.8%), tiny2 ([9, 12), 54.8%), tiny3 ([12, 20), 35.4%), and small ([20, 32], 1.9%). The category “tiny” denotes the union of tiny1-3. Mark [†] indicates that the method is modified to the version of the RGBT baseline detector. The number in bold indicates the best score.

Method	Backbone	mAP ₅₀ ^{all}	mAP ₅₀ ^{tiny}	mAP ₅₀ ^{tiny1}	mAP ₅₀ ^{tiny2}	mAP ₅₀ ^{tiny3}	mAP ₅₀ ^{small}	mAP ₂₅ ^{all}	FPS	Param	GFLOPs
Faster R-CNN [†]	ResNet50-FPN	26.81	27.06	0.00	20.78	34.06	36.00	40.11	29.3	65.07	125.05
Cascade R-CNN [†]	ResNet50-FPN	30.50	31.63	0.00	28.37	37.71	28.97	42.06	23.7	92.85	152.69
RetinaNet [†]	ResNet50-FPN	21.27	21.88	4.78	13.25	31.49	14.47	35.33	30.8	61.42	109.86
FCOS [†]	ResNet50-FPN	36.14	37.34	17.24	25.79	43.42	25.85	52.78	33.4	78.90	98.66
ATSS [†]	ResNet50-FPN	37.56	38.87	17.68	25.08	44.92	23.87	53.68	33.6	55.44	97.45
GFL [†]	ResNet50-FPN	39.04	40.83	18.55	28.58	47.47	28.29	54.53	33.5	55.89	99.66
SSPNet [†]	ResNet50-SSFPN	30.43	30.27	0.00	24.90	36.87	40.16	42.97	17.5	70.72	290.89
ATSS [†] w/NWD	ResNet50-FPN	35.07	36.62	16.98	24.66	42.64	22.97	52.45	33.2	55.44	97.45
FCOS [†] w/RFLA	ResNet50-FPN	36.80	37.90	24.04	27.69	43.33	23.15	53.68	33.5	78.73	97.52
CFT	CSPDarknet-PANet	23.81	24.40	4.73	14.71	33.39	17.43	40.12	25.1	147.74	123.07
HRFuser	HRFormer-HRFPN	25.06	25.87	0.00	30.92	30.24	22.81	35.43	15.6	48.81	91.61
Tlnet	ResNet50-FPN	26.84	26.61	0.00	25.73	33.32	33.67	39.88	22.7	99.56	137.28
QFDet	ResNet50-FPN	41.91	43.72	18.14	30.93	49.39	26.32	57.20	21.7	60.18	162.8
VTsArNet	Darknet53-Focus	40.08	41.47	17.82	28.82	47.69	26.91	56.52	16.9	49.96	112.6
IM-CMDet	ResNet50-FPN	41.73	43.22	18.47	29.00	49.79	27.60	58.12	20.3	109.74	166.94
CCRDet	ResNet50-FPN	43.58	45.14	29.53	35.69	50.62	33.06	58.34	23.6	71.82	123.44

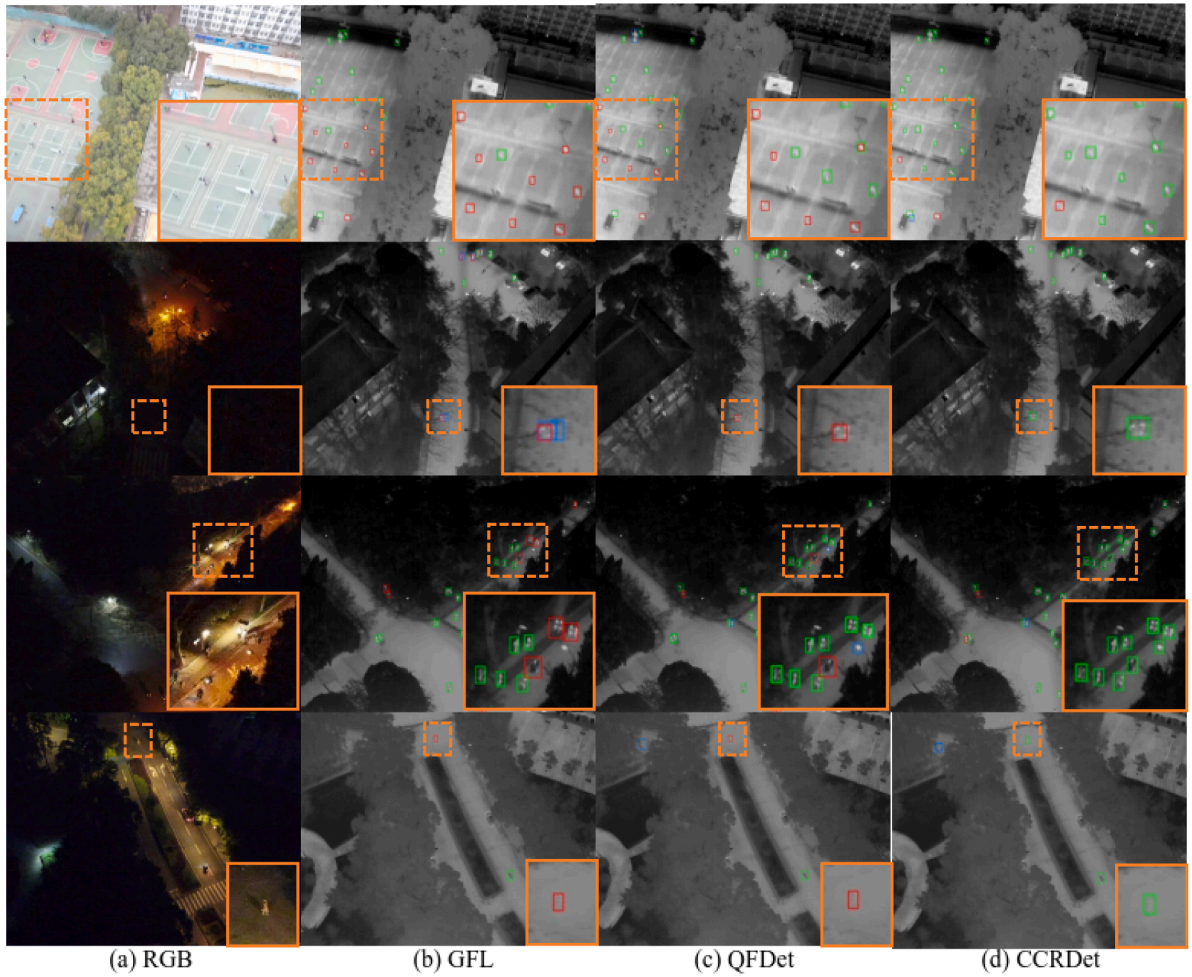


Fig. 4. Visualization of the proposed CCRDet and compared representative methods on the RGBTDronePerson dataset. From left to right are the RGB image, detection results of the baseline detector GFL, the second competitor QFDet, and the proposed CCRDet. The solid orange box provides a zoomed-in visualization of the area highlighted by the dashed orange box. Green, red, and blue boxes denote true positives, false negatives, and false positives, respectively.

an mAP₅₀^{tiny} of 40.25%. However, the proposed CCRDet outperforms the second competitor QFDet by 1.67% and 1.42% in mAP₅₀^{all} and mAP₅₀^{tiny}, respectively. The results demonstrate the superiority of the proposed CCRDet in effectively leveraging complementary information from RGB and thermal modalities for tiny person detection. Finally, as shown in

Table 1 columns 9–11, the proposed CCRDet achieves a trade-off between detection performance and computational complexity.

A visualization comparison between CCRDet, the second competitor QFDet, and the baseline detector GFL is shown in **Fig. 4**. The solid orange box provides a zoomed-in visualization of the area highlighted

Table 2

Performance comparison with the advanced methods on VTUAV-det. The three size categories are defined as follows: small ($[0, 32)$, 20.32%), medium ($[32, 96)$, 67.41%), and large ($[96, \infty)$, 12.27%). Mark [†] indicates that the method is modified to the version of the RGBT baseline detector. The number in bold indicates the best score.

Method	Backbone	mAP	mAP ₅₀	mAP ₇₅	mAP _s	mAP _m	mAP _l	FPS
Faster R-CNN [†]	ResNet50-FPN	26.6	59.5	19.7	1.6	28.6	52.8	25.48
Cascade R-CNN [†]	ResNet50-FPN	28.8	62.7	22.2	2.9	31.3	54.5	19.46
RetinaNet [†]	ResNet50-FPN	22.3	52.6	16.2	2.7	23.2	55.5	27.23
FCOS [†]	ResNet50-FPN	27.4	64.3	19.4	6.9	27.4	54.2	29.75
ATSS [†]	ResNet50-FPN	28.8	66.7	20.3	8.5	28.3	55.1	29.82
GFL [†]	ResNet50-FPN	29.8	67.8	22.1	9.5	29.9	56.3	29.4
SSPNet [†]	ResNet50-SSFPN	27.1	60.1	20.4	1.4	29.1	54.7	30.17
ATSS [†] w/NWD	ResNet50-FPN	28.0	65.4	20.2	7.2	27.7	55.8	29.82
FCOS [†] w/RFLA	ResNet50-FPN	27.4	66.1	17.8	9.5	27.9	51.9	15.89
CFT	CSPDarknet-PANet	24.2	58.6	16.7	0.5	28.4	48.3	20.79
HRFuser	HRFormer-HRFPN	27.3	59.8	21.1	3.8	29.5	53.6	13.09
Tinet	ResNet50-FPN	25.8	59.6	17.4	0.9	28.8	47.8	19.74
QFDet	ResNet50-FPN	31.4	69.8	21.7	11.9	29.7	56.2	19.2
VTsARNet	Darknet53-Focus	30.1	68.3	22.1	10.2	30.3	56.3	15.4
IM-CMDet	ResNet50-FPN	31.5	71.0	23.1	13.3	30.9	56.8	18.3
CCRDet	ResNet50-FPN	32.6	73.1	24.4	14.8	32.1	57.6	20.7

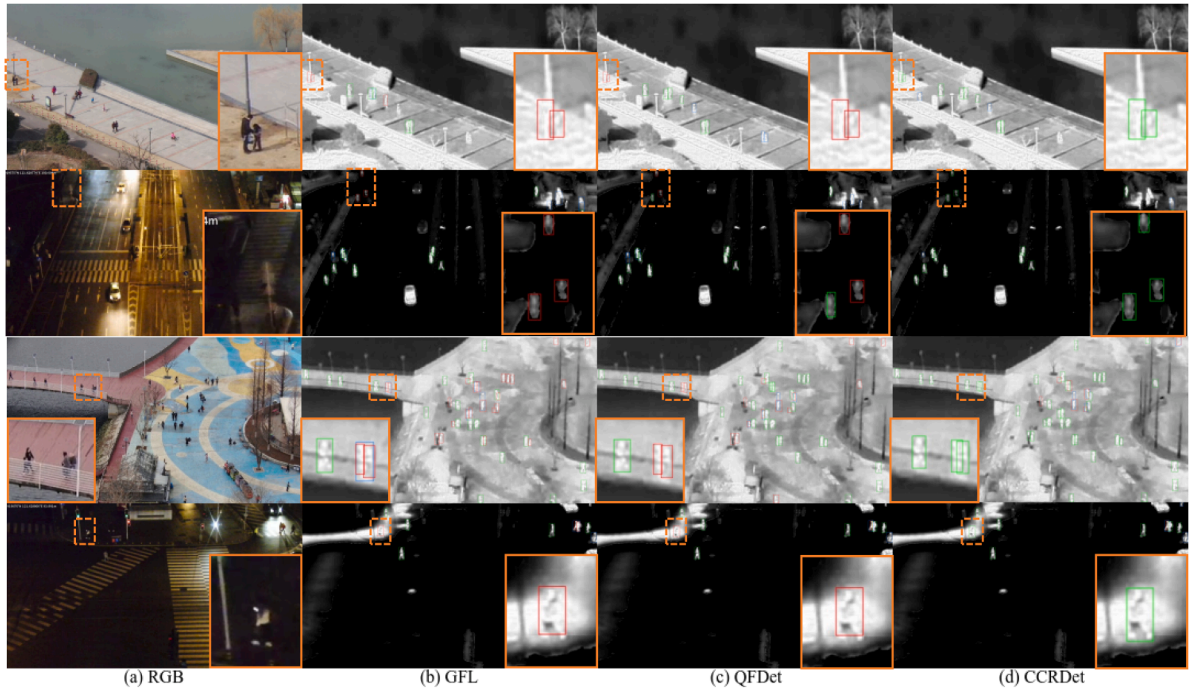


Fig. 5. Visualization of the proposed CCRDet and compared representative methods on the VTUAV-det dataset. From left to right are the RGB image, detection results of the baseline detector GFL, the second competitor QFDet, and the proposed CCRDet. The solid orange box provides a zoomed-in visualization of the area highlighted by the dashed orange box. Green, red, and blue boxes denote true positives, false negatives, and false positives, respectively.

by the dashed orange box. Green, red, and blue boxes denote true positives, false negatives, and false positives, respectively. For a comprehensive and effective comparison, four different scenarios were picked out, including daytime and nighttime. Comparing the above methods horizontally, it can be seen that the proposed method detects fewer false negatives in the four scenes.

In the first scene, thermal background clutter obscures pedestrian signatures, leading to more false negatives in GFL and QFDet. The proposed CCRDet is capable of recognizing background clutter regions in the thermal modality and guiding the model to effectively leverage complementary information from the corresponding regions in RGB, thereby enabling accurate pedestrian detection. In the second scene, poor illumination in RGB results in weak pedestrian signatures, leading to false negatives in GFL and QFDet. The proposed CCRDet manages to be aware of

Table 3

Ablation study of the proposed methods on RGBTDronePerson.

Model	CRG	MFGF	mAP ₅₀ ^{all}	mAP ₅₀ ^{tiny}	mAP ₅₀ ^{tiny1}	mAP ₅₀ ^{tiny2}	mAP ₅₀ ^{tiny3}
✗	✗	✗	39.04	40.83	18.55	28.58	47.47
✓	✗	✗	42.59	44.38	28.63	31.55	49.54
✗	✓	✓	41.11	42.93	25.91	31.17	49.01
✓	✓	✓	43.58	45.14	29.53	35.69	50.62

the poorly illuminated regions in RGB and effectively leverages complementary information from thermal, thereby successfully detecting the pedestrians. These indicate that the proposed method is more effective in addressing these challenging regions.

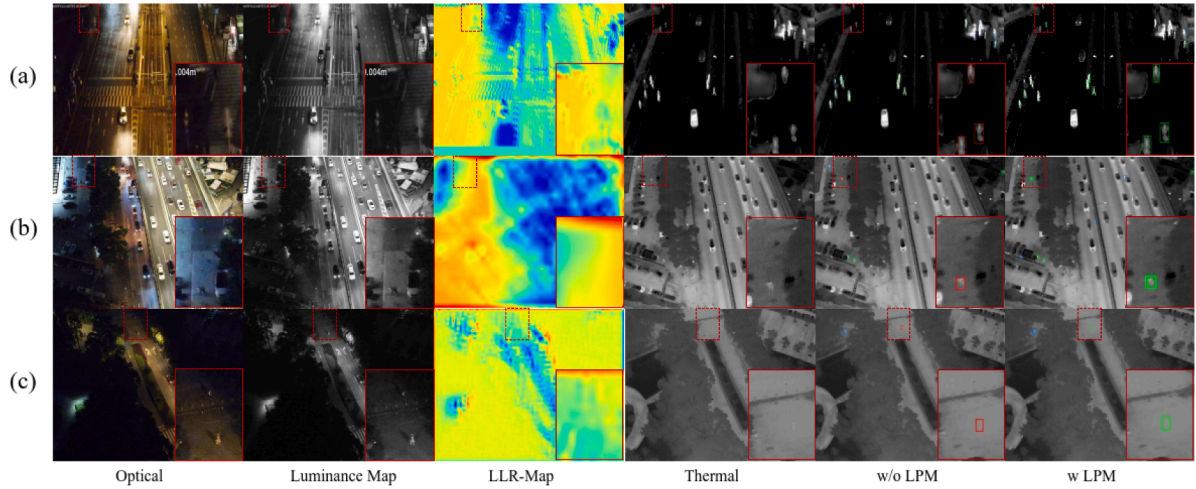


Fig. 6. Examples of low-light region-aware maps (LLR-Map) generated by LPM. From left to right, each column represents the RGB image, the Luminance Map, the LLR-Map, the thermal image, detection results of the model without LPM, and detection results of the model with LPM, respectively.

Table 4

Ablation study of the proposed LPM and BCAM on RGBTDronePerson.

Model		mAP_{50}^{all}	mAP_{50}^{tiny}	mAP_{50}^{tiny1}	mAP_{50}^{tiny2}	mAP_{50}^{tiny3}
LPM	BCAM					
$\times$	$\times$	39.04	40.83	18.55	28.58	47.47
$\checkmark$	$\times$	40.70	42.37	19.04	30.73	48.11
$\times$	$\checkmark$	41.55	43.63	27.97	31.21	48.67
$\checkmark$	$\checkmark$	42.59	44.38	28.63	31.55	49.54

On VTUAV-det. As shown in the first row of Table 2, “s”, “m”, and “l” denote (small, 20.32%), (medium, 67.41%), and (large, 12.27%) objects, respectively. As shown in Table 2 columns 3 and 4, the proposed CCRDet achieved an mAP of 32.6% and an mAP_{50} of 73.1%. Compared to these representative methods, CCRDet achieves substantial improvements with increases of 1.1% and 2.1% in mAP and mAP_{50} , respectively. As shown in Table 2 columns 6–8, the proposed CCRDet achieves improvements of 1.3%, 1.5%, and 1.2% in mAP_s , mAP_m , and mAP_l , respectively, demonstrating the effectiveness of the proposed methods in detecting objects of various sizes, especially small objects.

A visualization comparison between CCRDet, QFDet, and the baseline detector GFL is shown in Fig. 5. Four different scenarios were picked out, including daytime and nighttime. Comparing the above methods horizontally, it can be seen that the proposed method detects fewer false negatives of the four scenes. This further substantiates the effectiveness of the proposed CCRDet in addressing poorly illuminated regions in RGB and background clutter regions in thermal.

4.4. Ablation study

To assess the effectiveness of the proposed CRG and MFGF, ablation experiments were conducted on the RGBTDronePerson dataset. The baseline was a two-stream GFL, where RGB and thermal features were extracted by separate backbones and fused via concatenation with a 1×1 convolution. The quantitative results are shown in Table 3.

From Table 3 rows 2 and 3, adopting CRG can provide 3.55% improvement in mAP_{50}^{all} . In detail, for tiny1, tiny2, and tiny3 categories, it can provide 10.08%, 2.97%, and 2.07% improvements in mAP_{50}^{tiny1} , mAP_{50}^{tiny2} , and mAP_{50}^{tiny3} , respectively. This indicates that the CRG is particularly effective in handling extremely tiny objects, which are more prevalent in the tiny1 category, thereby demonstrating its substantial capability in enhancing valid target features during feature extraction. From Table 3 rows 2 and 4, adopting MFGF can provide 2.07% improvement in mAP_{50}^{all} . This suggests that it can function effectively on its own.

More interestingly, when both CRG and MFGF are employed, the performance in mAP_{50}^{tiny2} improves by 7.11% compared to the baseline. When adopting only CRG or MFGF, the improvements in mAP_{50}^{tiny2} are only 2.97% and 2.59%, respectively. It is worth noting that tiny2 accounts for 54.8% of the total instances, this indicates that CRG and MFGF can reinforce each other, and their combination can yield more significant improvements.

4.5. Analysis of cross-modal region-aware guidance

To validate the effectiveness of the proposed LPM and BCAM in CRG, ablation experiments were conducted on the RGBTDronePerson dataset. The quantitative results are shown in Table 4. From Table 4 rows 2 and 3, adopting LPM can provide 1.66% improvement in mAP_{50}^{all} . In detail, for tiny1, tiny2, and tiny3 categories, it can provide 0.49%, 2.15%, and 0.64% improvements in mAP_{50}^{tiny1} , mAP_{50}^{tiny2} , and mAP_{50}^{tiny3} , respectively. This suggests that it can function effectively on its own. From Table 4 rows 2 and 4, adopting BCAM can provide 2.51% improvement in mAP_{50}^{all} . In detail, for tiny1, tiny2, and tiny3 categories, it can provide 9.42%, 2.63%, and 1.20% improvements in mAP_{50}^{tiny1} , mAP_{50}^{tiny2} , and mAP_{50}^{tiny3} , respectively. This suggests that it can function effectively on its own. Meanwhile, it is particularly effective in handling extremely tiny objects.

To further reveal the mechanism of the proposed LPM, the low-light region-aware maps (LLR-Map) are visualized. As shown in Fig. 6, three representative examples encompassing distinct low-light regions are presented. For each example, from left to right, each column represents the RGB image, the Luminance Map, the LLR-Map, the thermal image, detection results of the model without LPM, and detection results of the model with LPM, respectively. The solid red box provides a zoomed-in visualization of the area highlighted by the dashed red box. As illustrated in columns 2 and 3 of Fig. 6, the LLR-Map generated by LPM effectively highlights low-light regions in the luminance map. From columns 5 and 6 of Fig. 6, it can be observed that in these low-light regions, the model with LPM successfully detects the pedestrians, while the model without LPM fails. This indicates that LPM enables the model to be aware of low-light regions in RGB and effectively leverage complementary information in the counterpart regions from thermal.

To further reveal the mechanism of the proposed BCAM, the background clutter region-aware maps (BCR-Map) are visualized. As shown in Fig. 7, three representative examples encompassing distinct background clutter regions are presented. For each example, from left to right, each column represents the thermal image, the BCR-Map, the RGB image, the detection results of the model without BCAM, and the

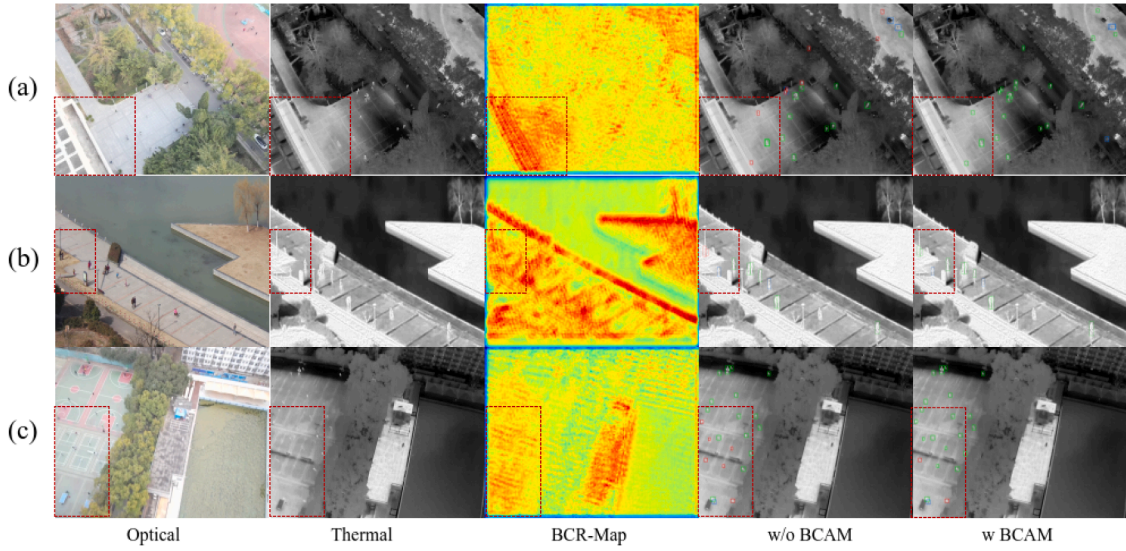


Fig. 7. Examples of background clutter region-aware maps (BCR-Map) generated by BCAM. From left to right, each column represents the RGB image, the BCR-Map, the thermal image, detection results of the model without BCAM, and detection results of the model with BCAM, respectively.

Table 5

Comparative experiment between explicit Y-channel extraction and feature extraction from RGB on RGBTDronePerson.

Method	mAP_{50}^{all}	mAP_{50}^{tiny}	mAP_{50}^{tiny1}	mAP_{50}^{tiny2}	mAP_{50}^{tiny3}
Baseline	39.04	40.83	18.55	28.58	47.47
Feature extraction from RGB	39.75	40.22	12.31	30.20	45.34
Explicit Y-channel extraction	40.70	42.37	19.04	30.73	48.11

Table 6

Comparative experiment between RGB-to-YUV (extracting the Y channel) and RGB-to-LAB (extracting the L channel) on the RGBT-DronePerson.

Method	mAP_{50}^{all}	mAP_{50}^{tiny}	mAP_{50}^{tiny1}	mAP_{50}^{tiny2}	mAP_{50}^{tiny3}
RGB-to-LAB	40.15	41.23	14.96	30.12	46.26
RGB-to-YUV	40.70	42.37	19.04	30.73	48.11

detection results of the model with BCAM, respectively. Key areas are marked with a dashed red box. As illustrated in columns 1 and 2 of Fig. 7, the BCR-Map generated by BCAM effectively highlights background clutter regions in thermal images. From columns 4 and 5 of Fig. 7, it can be observed that in these background clutter regions, the model with BCAM detects fewer false negatives than the model without BCAM. This indicates that BCAM enables the model to be aware of background clutter regions in thermal and effectively leverage complementary information in the counterpart regions from RGB.

The Y channel in the YUV color space provides a physically meaningful luminance signal that closely aligns with both human visual perception and camera sensor responses. By explicitly extracting the Y channel, a clear luminance prior is introduced instead of requiring the backbone network to implicitly disentangle illumination information from the coupled RGB channels. This design helps alleviate the optimization burden and improves robustness under challenging lighting conditions [29].

To demonstrate the superiority of explicit Y-channel extraction over learned feature representations, additional comparative experiments have been conducted. In the Luminance Perception Module (LPM), the Luminance Map derived from explicit Y-channel extraction was replaced with a convolutional transformation applied directly to the RGB input, while all other components were kept unchanged. LPM was used separately on the Baseline GFL[†]. The corresponding experimental results are presented in Table 5.

As shown in Table 5, on the RGBTDronePerson dataset, the Explicit Y-channel extraction approach consistently achieves the best performance across all key metrics. Compared to the baseline, explicit Y-channel extraction improves the overall mAP_{50}^{all} from 39.04% to 40.70% (an absolute gain of +1.66%). When compared directly to Feature extraction from RGB, the advantage of explicit Y-channel extraction becomes more significant. The overall mAP_{50}^{all} is higher by approximately 0.95 points, while the gap is particularly striking on the most challenging extremely small object, tiny1: 19.04% versus only 12.31%. This indicates that implicit luminance learning from RGB channels is more susceptible to color noise and illumination variations, whereas explicit separation of the Y channel offers a more robust and stable brightness representation.

To further validate the effectiveness of the proposed strategy in comparison with the RGB-to-LAB conversion adopted in M2D-LIF [28], comparative experiments have been conducted between YUV-based and LAB-based illumination extraction within the same framework. The corresponding experimental results are presented in Table 6.

As shown in Table 6, on the RGBTDronePerson dataset, the RGB-to-YUV approach consistently achieves better performance across all key metrics compared to the RGB-to-LAB approach. The overall mAP_{50}^{all} is higher by approximately 0.55 points, while the gap is particularly notable on the extremely small object tiny1: 19.04% versus only 14.96%. This indicates that the YUV color space, which is designed to separate luminance from chrominance, is more effective for luminance extraction in the context of low-light region awareness, whereas the LAB color space, which is designed to be perceptually uniform, may not provide as distinct a separation between luminance and chrominance for this specific application.

4.6. Analysis of modality-difference feature gated fusion

To reveal the mechanism of the proposed MFGF, heat maps of the RGB, thermal, and fused features are visualized. As shown in Fig. 8,

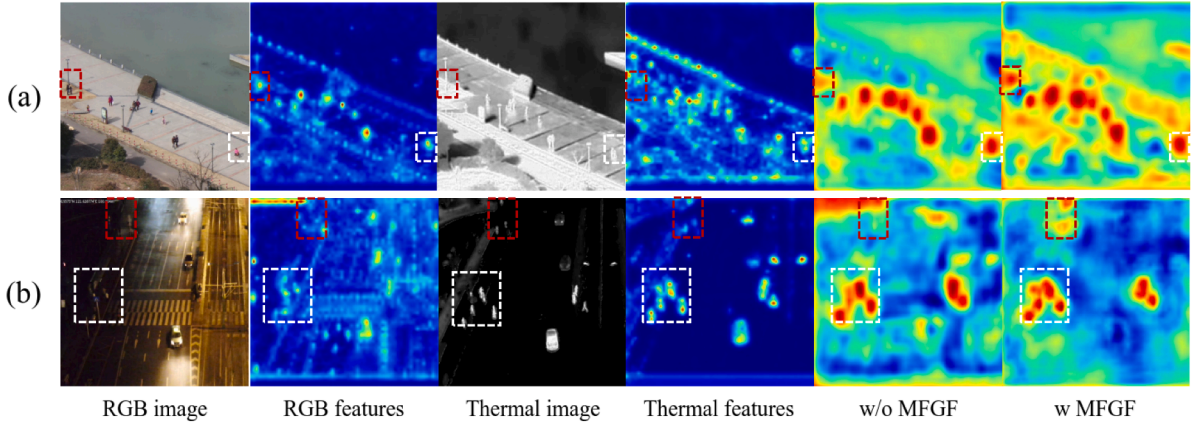


Fig. 8. Examples of heat maps corresponding to RGB, thermal, and fused features. From left to right, each column represents the RGB image, the heat map of the RGB features, the thermal image, the heat map of the thermal features, the heat map of the fused features via concatenation with a 1×1 convolution, and the heat map of the fused features with MFGF, respectively.

two representative scenarios are presented. For each scenario, from left to right, each column represents the RGB image, the RGB features heat map, the thermal image, the thermal features heat map, the features fused via concatenation with a 1×1 convolution, and the features fused via the proposed MFGF, respectively. Key areas are marked with a dashed box. The red dashed boxes indicate objects located in poorly illuminated RGB regions or thermally cluttered backgrounds. The white dashed boxes denote targets that are simultaneously captured by both modalities.

In the first scenario, the pedestrian is well illuminated in RGB, while the thermal image is affected by background clutter. As shown in columns 2 and 4 of Fig. 8, the RGB features heat map effectively highlights the pedestrian, while the thermal features heat map is distracted by the background clutter. As shown in columns 5 and 6 of Fig. 8, the features fused via the proposed MFGF effectively highlight the pedestrian, the response is significantly stronger than that of the features fused without MFGF. In the second scenario, the pedestrian is poorly illuminated in RGB, while the thermal image is less affected by background clutter. As shown in columns 2 and 4 of Fig. 8, the RGB features heat map fails to highlight the pedestrian, while the thermal features heat map effectively highlights the pedestrian. As shown in columns 5 and 6 of Fig. 8, the features fused via the proposed MFGF effectively highlight the pedestrian, while the features fused without MFGF fail. These indicate that the proposed MFGF delivers valid target features to the fused features with effective preservation, thereby enhancing their response intensity after fusion.

In the MFGF module, the subtraction operation serves only as an intermediate step for weight estimation during the fusion process. The final feature integration ultimately reduces to a more robust additive fusion mechanism. The MFGF module utilizes subtraction to obtain the weighting coefficients. If both modalities simultaneously capture the target, the subtraction operation yields similar weighting coefficients for the two modalities, and the valid target features detected by both modalities are ultimately fused in a manner approximating weighted summation.

As shown in columns 2, 4, and 6 of Fig. 8, both modalities captured the target simultaneously, and the features fused by the MFGF module exhibit strong response intensity to the targets. This indicates that when both modalities capture the target simultaneously, the MFGF module does not cause significant suppression of the target signal itself, while still ensuring effective fusion performance.

Table 7

Real-time performance on NVIDIA Jetson AGX Xavier for the RGBTDronePerson Dataset. GFL[†] denotes the baseline. Best results are highlighted in **bold**. And the second one is marked with underline.

Method	Backbone	mAP ₅₀ ^{all}	FPS↑	Inf. (ms)↓
GFL [†]	ResNet50-FPN	39.04	16.5	60.6
QFDet	ResNet50-FPN	<u>41.91</u>	12.1	82.6
VTsARNet	Darknet53-Focus	40.08	7.8	128.2
CCRDet	ResNet50-FPN	43.58	<u>13.7</u>	<u>72.5</u>

4.7. Real-time performance on embedded platforms

To evaluate the real-time performance of the proposed CCRDet on embedded devices, the proposed method, baseline, and two advanced fusion-based methods were deployed on the NVIDIA Jetson AGX Xavier, a widely used edge AI computing platform. The inference speed was measured in frames per second (FPS) and inference time (ms).

As shown in Table 7, the proposed CCRDet achieves an inference speed of 13.7 FPS with an inference time of 72.5 ms on the NVIDIA Jetson AGX Xavier. Compared with the baseline GFL[†], CCRDet maintains a comparable inference speed while achieving a significant improvement of 4.54% in mAP₅₀^{all}. Compared with advanced fusion-based methods, CCRDet exhibits superior performance and demonstrates an advantage in inference efficiency, reducing the inference time by 10.1 ms and 55.7 ms compared with QFDet and VTsARNet, respectively. Nevertheless, the inference efficiency of the proposed method can be further improved, and a more lightweight design will be explored for optimization in future work.

4.8. Failure case

Despite the effectiveness of the proposed CCRDet, certain challenging scenarios remain unresolved. As shown in Fig. 9, two representative failure cases are presented. For each case, from left to right, each column represents the RGB image, the detection results of the competitive method QFDet, and the detection results of the proposed CCRDet, respectively. Key areas are marked with a dashed red box.

In the first case, the interfering heat sources in the thermal image lead to numerous false positives in both QFDet and the proposed

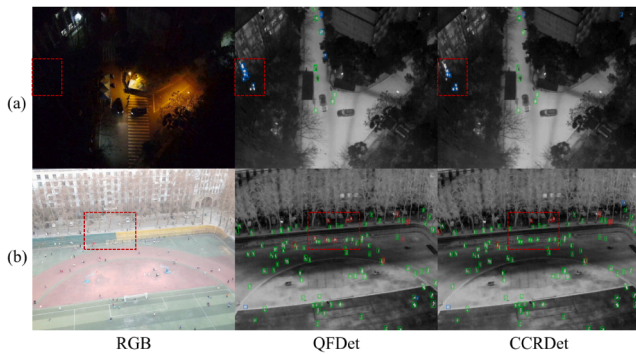


Fig. 9. Examples of failure cases. From left to right, each column represents the RGB image, the detection results of the competitive method QFDet, and the detection results of the proposed CCRDet, respectively. Green, red, and blue boxes denote true positives, false negatives, and false positives, respectively.

CCRDet. This demonstrates that the model fails to discriminate interfering heat sources that exhibit similar thermal responses to the target. In the second case, the pedestrian appears extremely small and partially occluded in the RGB image, while the surrounding background in the infrared image is complex. When the target quality in both modalities is degraded, existing methods still inevitably suffer from missed detections. These limitations indicate potential directions for future improvements.

5. Conclusion

In this article, a CCRDet has been proposed for effective drone-based RGBT tiny person detection under poorly illuminated RGB regions and thermally cluttered backgrounds. CCRDet employs the proposed cross-modal region-aware guidance to generate region-aware maps and guide the counterpart modality to enhance valid target features accordingly. After that, it leverages the proposed modality-difference feature gated fusion to deliver valid target features to the fused features with effective preservation, thereby enhancing their response intensity after fusion and providing high-quality inputs for the detection head.

The quantitative experimental results evaluated on two datasets demonstrate the effectiveness and competitiveness of the proposed CCRDet. The visualization of the region-aware maps demonstrates that the proposed CRG can enable the model to be aware of poorly illuminated regions in RGB and background clutter regions in thermal. Meanwhile, the visualization of the feature heat maps demonstrates that the proposed MFGF can deliver valid target features to the fused features with effective preservation.

Based on the failure cases observed in the experimental results, the current limitations of the proposed CCRDet are further analyzed and clarified. First, the method lacks sufficient capability to discriminate interfering heat sources that exhibit target-like thermal responses, which may lead to false positives. Second, missed detections still occur when the target quality in both modalities is simultaneously degraded. To address these limitations, several future research directions are envisioned, including the development of more effective strategies for distinguishing interfering heat sources and improving the model's robustness under severely degraded cross-modal target conditions.

CCRediT authorship contribution statement

Panzhe Gu: Writing – original draft, Visualization, Software, Methodology; **Jue Wang:** Writing – review & editing, Conceptualization; **Tianyu Wei:** Writing – review & editing; **Guoqing Wang:** Writing – review & editing; **Liang Chen:** Supervision; **Wenchao Liu:** Validation, Project administration.

Data availability

The data used in this study are publicly available.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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